This illustration suggests that the mean or expected value of any random variable may be obtained by the sum of the product of each value of the random variable by its corresponding probability.

So average $= \sum$ (value of x) $\times$ (probability)

This is true if the random variable is discrete. In the case of continuous random variable, the mathematical expectation is essentially the same with summations being replaced by integrals.

Two quantities are often used to summarize a probability distribution of a random variable X. In terms of statistics one is central tendency and the other is dispersion or variability of the probability distribution. The mean is a measure of the centre tendency of the probability distribution, and the variance is a measure of the dispersion, or variability in the distribution. But these two measures do not uniquely identify a probability distribution. That is, two different distributions can have the same mean and variance. Still, these measures are simple, and useful in the study of the probability distribution of X.

11.5.1 Mean

Definition 11.8 : (Mean)

Suppose X is a random variable with probability mass (or) density function $f(x)$. ***The expected value or mean or mathematical expectation of X*** **,** denoted by $E(X)$ or μ *is*

$$E(X) = \begin{cases} \sum_x x\, f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x\, f(x)\, dx & \text{if } X \text{ is continuous} \end{cases}$$

The expected value is in general not a typical value that the random variable can take on. It is often helpful to interpret the expected value of a random variable as the long-run average value of the variable over many independent repetitions of an experiment.

Theorem 11.3 (Without proof)

Suppose X is a random variable with probability mass (or) density function $f(x)$. ***The expected value of the function*** $g(X)$, *a new random variable is*

$$E(g(X)) = \begin{cases} \sum_x g(x)\, f(x) & \text{if } g(x) \text{ is discrete} \\ \int_{-\infty}^{\infty} g(x)\, f(x)\, dx & \text{if } g(x) \text{ is continuous} \end{cases}$$

If $g(X) = x^k$ the above theorem yield the expected value called *the k-th moment about the origin* of the random variable *X*.

Therefore the k-th moment about the origin of the random variable *X* is

$$E(X^k) = \begin{cases} \sum_x x^k\, f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x^k\, f(x)\, dx & \text{if } X \text{ is continuous} \end{cases}$$

12th_Maths_Vol 2_EM_CH 11_Probability Distributions_29-05-20.indd 204 16-12-2022 16:51:43